

ATHLETEOS

Personalized Gym Athlete Web App

PRODUCT REQUIREMENTS DOCUMENT (PRD)

PRODUCT THESIS

One athlete profile that continuously converts body metrics, training performance, nutrition logs, habits, recovery signals and optional computer-vision measurements into a practical daily plan.

Working product name	AthleteOS
Document version	1.0
Prepared for	Personalized gym / fitness web application
Primary platforms	Responsive web + installable PWA; mobile-first
Date	11 August 2026

Bodybuilding • Calisthenics • Athletic Performance • Aesthetic Physique • Hybrid

00 Document Map

What this PRD contains

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Executive summary

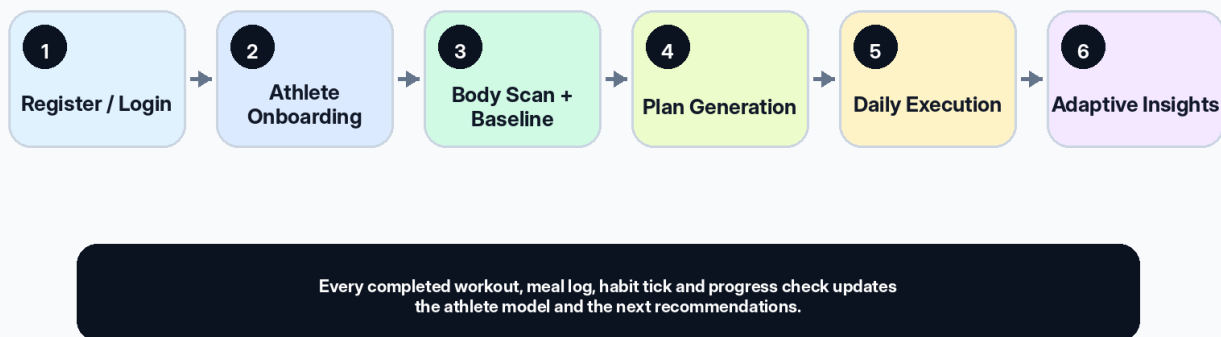
AthleteOS is a personalized fitness operating system for gym athletes whose training goal may be bodybuilding, calisthenics, athletic performance, aesthetic physique development, or a hybrid. Unlike a simple workout logger, the product maintains an evolving athlete profile and uses that profile to generate daily actions, track execution, and explain progress over time.

The initial product should prioritize reliable logging and useful adaptation over “AI for AI’s sake.” The strongest personalization signals are workout history, performance progression, recovery, bodyweight trend, nutrition adherence, equipment, schedule and user goals. Computer vision is an assistive signal for standardized progress assessment, posture/pose quality and visual proportion tracking; it must not be presented as a medical body-fat scanner or diagnosis.

CORE PRODUCT LOOP

Assess → Plan → Execute → Log → Analyze → Adapt. Every feature should strengthen this loop rather than create isolated dashboards.

AthleteOS Core User Flow



01 Product Vision & Scope

A single personal fitness system that grows with the athlete

1.1 Product vision

Build a web app that understands where an athlete is now, what physique/performance outcome they want, what they can realistically do each week, and how their body and performance are responding. The app should then turn that information into a daily training, nutrition, hydration, habit and recovery workflow.

1.2 Product goals

- Personalize training by athlete type, training level, equipment, schedule, performance history and recovery.
- Make workout logging fast enough to use between sets and detailed enough for meaningful progression analysis.
- Make food logging practical for Indian users while retaining global food coverage and packaged-food support.
- Create a privacy-conscious body assessment workflow using computer vision for pose, proportions and repeatable progress comparison.
- Convert raw logs into week-over-week and month-over-month insights that tell the user what is improving, stagnant or declining.
- Encourage consistency through streaks, habit grids, milestones and low-friction daily check-ins.

1.3 Non-goals for MVP

- Medical diagnosis, injury diagnosis or treatment recommendations.
- Claiming accurate body-fat percentage from a single photo.
- Replacing a registered dietitian, physician, physiotherapist or certified coach for high-risk cases.
- Fully autonomous exercise technique grading for every movement on day one.
- A public social network or marketplace before the core tracking loop is proven.

1.4 Product principles

Principle	Product implication
Action before analytics	Today screen shows the next useful action before historical charts.
Progressive disclosure	Beginners see simple prompts; advanced users can log RIR/RPE, tempo, volume and detailed metrics.
Explain recommendations	"Why this changed" is shown whenever the system modifies volume, load, calories or exercise selection.
Privacy by default	Photo processing can be on-device; storing raw images is optional and explicitly consented.
No fake precision	Nutrition, body measurements and readiness are labeled as estimates where appropriate.
Flexible athlete identity	Users can be 70% bodybuilding + 30% calisthenics instead of being locked to one mode.

02 Users, Goals & Athlete Modes

Who the product serves and how personalization begins

2.1 Primary athlete personas

Persona	Primary objective	What they need most
Bodybuilding athlete	Muscle hypertrophy, symmetry, weak-point development	Volume progression, muscle-group frequency, exercise history, bodyweight/nutrition trend.
Calisthenics athlete	Strength-to-weight ratio and skills	Skill prerequisites, holds/time, assistance level, bodyweight reps, progression trees.
Athletic athlete	Speed, power, strength, conditioning	Performance tests, sprint/jump/conditioning logs, fatigue management, mobility.
Aesthetic physique athlete	Balanced proportions, leanness, V-taper /	Visual progress, hypertrophy plan,

Persona	Primary objective	What they need most
	lower-body balance	adherence, body measurements, posing/photo consistency.
Hybrid athlete	Combination of strength, physique and skills	Weighted priorities and conflict-aware weekly scheduling.

2.2 Training level taxonomy

Level	Suggested interpretation	System behavior
Beginner	New to structured training / technique still being learned	Low complexity, stable exercise selection, conservative progression, technique education.
Early Beginner	Basic movement patterns established; building consistency	Simple double progression, slightly more volume, first specialization blocks.
Intermediate	Consistent training history with meaningful plateaus	Periodized volume/intensity, weak-point analysis, more autoregulation.
Advanced	High training age and specialized goals	Fine-grained load management, advanced exercise rotation, individualized fatigue thresholds.

IMPORTANT

Do not rely only on the user's self-selected level. Create a derived "system level" from training age, recent consistency, key movement performance, skill competency and available history. The user can still override it.

2.3 Goal model

- Primary: muscle gain, fat loss, body recomposition, strength, calisthenics skill, athletic performance, conditioning, general fitness.
- Secondary physique focus: shoulders, chest, back width, back thickness, arms, glutes, quads, hamstrings, calves, core.
- Priority weighting: user can assign percentages across 2-3 goals; total = 100%.
- Constraint inputs: days/week, session duration, training location, equipment, preferred/avoided exercises, past pain/injuries, dietary preference, budget and meal pattern.

03 End-to-End User Journey

From registration to a continuously adapting athlete profile

3.1 First-time onboarding flow

1. Create account: email/phone/social sign-in; verify contact method; accept Terms and Privacy Notice.
2. Choose athlete mode: bodybuilding, calisthenics, athletic, aesthetic, or hybrid; choose primary goal.
3. Baseline profile: age range/DOB, preferred units, height, weight, optional waist/chest/hip/arm/thigh measurements.
4. Daily baseline: current water intake, sleep duration, step/activity level, training days, typical session length.
5. Experience: self-selected level + training age + recent consistency + optional competency questions.
6. Equipment: commercial gym / home gym / bodyweight / selected equipment inventory.
7. Nutrition preferences: vegetarian/eggetarian/vegan/non-vegetarian, allergies, disliked foods, meal frequency, cuisine preferences.
8. Optional body scan: consent → capture guide → image quality check → pose/segmentation → proportions → review result.
9. Plan preview: weekly training structure, nutrition targets/behaviors, habits, baseline metrics and "why this plan".
10. First Today screen: first workout, meals to log, water target/goal, habits and quick readiness check.

3.2 Returning daily flow

- Open Today → see workout, habits, nutrition status and recovery prompt.
- Start workout → log each set with reps/load or time/distance/assistance depending on exercise type.
- Log meals via search, recent foods, favorites, recipe, barcode or quick-add macros.
- Tick habits directly in grid or Today card.
- Finish day → short check-in: session difficulty, soreness, sleep plan and optional notes.

- System recalculates trends and only changes the plan when enough evidence exists.

04 Functional Requirements

Detailed product behavior by module

4.1 Authentication & account

ID	Requirement	Priority
AUTH-01	Register and sign in with email/password; add Google/Apple/phone as optional providers.	P0
AUTH-02	Email/phone verification, password reset, secure session management and device logout.	P0
AUTH-03	Profile export, account deletion and consent controls.	P0
AUTH-04	Optional passkeys / MFA for enhanced security.	P1
AUTH-05	Roles: athlete, coach (future), admin/support.	P1

4.2 Athlete profile & baseline metrics

- Required: height, weight, training goal, athlete mode, experience level, available days/week, equipment context.
- Recommended: age, waist, sleep, activity/steps, current water intake, dietary preference, session length.
- Optional: chest/hip/neck/arms/thighs/calves, resting heart rate, progress photos, performance tests, wearable sync.
- Metric history: all changing metrics must be timestamped instead of overwriting previous values.
- Unit system: kg/lb, cm/in; conversions must be lossless and consistent.

4.3 Daily Today screen

Card	Contents	Primary action
Training	Session name, estimated duration, completion %, readiness adjustment	Start / resume workout
Nutrition	Calories/macros or simplified meal adherence, remaining targets	Log meal
Hydration	Current vs chosen target	+ glass / +250 ml / custom
Habits	Today's habits + streaks	Tick complete
Recovery	Sleep, soreness, energy, optional resting HR	30-second check-in
Progress	Recent PR, weight trend, streak milestone	View insight

4.4 Streak system

Maintain separate streaks for workout adherence, meal logging, hydration logging, habits and an optional "overall consistency" streak. Avoid punishing rest days: a scheduled rest day counts as adherence when the user completes the planned recovery/check-in behavior.

05 Computer Vision Body Assessment

A useful body-structure feature without unsafe precision

5.1 Recommended CV stack

Use Google MediaPipe Pose Landmarker as the default web-first model. It can return body landmarks in image coordinates and 3D world coordinates; its web configuration can also output a segmentation mask. This makes it suitable for posture/pose validation, geometric ratios and consistent progress-photo analysis in the browser. [1]

Component	Recommended approach	Purpose
Pose landmarks	MediaPipe Pose Landmarker	Shoulders, hips, elbows, wrists,

Component	Recommended approach	Purpose
		knees/ankles when visible; pose normalization and symmetry features.
Person mask	Pose Landmarker segmentation mask or separate person segmentation when needed	Estimate silhouette boundaries and remove background from proportion calculations.
Quality gate	Rules + landmark confidence + image metadata	Reject rotated, cropped, poorly lit or highly angled captures.
Derived feature engine	Deterministic geometry first; ML later	Shoulder-to-hip visual ratio, torso/limb ratios, left/right alignment, progress deltas.
Optional advanced model	Custom regression/classifier trained on consented labeled data	Only after a validated dataset exists; never invent labels from weak ground truth.

5.2 Capture protocol

- Ask for front and side image at MVP; add back image later for better symmetry/progress comparison.
- Allow shirtless or fitted clothing. Loose clothing lowers confidence and should be clearly flagged.
- Neutral stance, arms slightly away from torso, camera approximately chest height, consistent distance, uncluttered background and even lighting.
- Display silhouette overlay so the user can align their body before capture.
- For progress comparison, guide the user to repeat the same camera angle, lighting and distance.

5.3 CV outputs shown to the user

Output	Example	Use in product
Capture quality score	92 / 100	Decide whether scan is valid for comparison.
Pose alignment	Shoulders level; slight trunk rotation detected	Ask for recapture or lower confidence.
Visual proportions	Shoulder-to-hip visual ratio, torso-to-leg proxy when full body available	Track directional change; help choose emphasis blocks.
Symmetry proxy	Left/right shoulder height or arm silhouette difference	Progress tracking; not diagnosis.
Standardized silhouette	Masked outline with normalized scale	Before/after overlay.
Confidence	High / medium / low by metric	Prevent false precision.

5.4 What the model must NOT claim

- Exact body-fat percentage from a normal photo.
- Medical posture diagnosis, scoliosis diagnosis or injury detection.
- “Ectomorph/mesomorph/endomorph” as a scientifically definitive body type.
- Genetic potential, hormone status or health condition inference from appearance.

PRIVACY-FIRST OPTION

Run pose/segmentation in the browser and store only derived landmarks/ratios unless the athlete explicitly enables progress-photo storage. This greatly reduces sensitivity and storage risk.

5.5 Body-structure personalization logic

The CV result should influence training only as a low-to-medium-weight signal. Example: if the athlete's goal is an aesthetic V-taper and the visual shoulder/hip ratio is below their own target trend, the planner may increase emphasis on lateral delts and lats if recovery and volume data permit. Performance history and user preference remain more important than appearance-derived features.

06 Nutrition & Food Dataset

Indian-first search with global coverage

6.1 Nutrition product goals

- Find a common Indian dish in 1-2 seconds, including spelling variants and transliterations.
- Support raw ingredients, home-cooked dishes, restaurant estimates, packaged products and user recipes.
- Show portion units users actually understand: katori, roti, piece, glass, cup, bowl, tablespoon, gram, ml.
- Calculate recipe nutrition from ingredients when a composite dish has no reliable direct nutrient record.
- Make “recent”, “frequent” and “favorites” faster than repeated search.

6.2 Recommended food data sources

Source	Best use	Implementation note
ICMR-NIN Indian Food Composition Tables 2017	Indian raw foods and foundational nutrient composition	Official NIN material reports 528 key foods and extensive nutrient/component coverage; use as authoritative Indian foundation where licensing/usage permits. [2]
USDA FoodData Central	Global foundational foods and nutrient normalization	REST API supports search/details; data are published as public-domain/CC0. Cache and normalize IDs. [3]
Open Food Facts	Packaged foods and barcode scanning across markets	Use product/barcode API for branded packaged foods; preserve source attribution and data-quality flags. [4]
AthleteOS recipe layer	Indian dishes + custom household recipes	Ingredients × edible quantity → computed nutrients; allow region/cooking-method variants.

6.3 Food entity schema

Field	Example / behavior
food_id	Stable internal UUID
canonical_name	Bhindi Sabzi
aliases	bhindi, okra sabzi, ladyfinger sabzi, bhindi ki sabji
language_aliases	Hindi/English/transliteration tokens
food_type	ingredient / recipe / packaged / restaurant / custom
cuisine_region	North Indian / South Indian / Mediterranean / etc.
diet_tags	vegetarian, vegan, eggetarian, halal-friendly metadata if sourced
serving_units	100 g, 1 katori, 1 cup, 1 piece
nutrients_per_100g	energy, protein, carbs, fat, fiber, sugar, sodium + optional micronutrients
source	IFCT / FDC / OFF / recipe-derived / user
confidence	verified / estimated / user-entered
ingredients	for recipes: linked ingredient IDs + quantity + yield

6.4 Search and matching requirements

- Fuzzy spelling: “aloo matar”, “alu mutter”, “alumatar”, “aalu matar” should converge.
- Synonym expansion: “ladyfinger” → okra → bhindi; “cottage cheese” → paneer where context matches.
- Cuisine/ranking boost based on user locale, history and diet preference without hiding global results.
- Search by Hindi transliteration and later native scripts.
- Result cards show serving, calories and macros plus source/confidence; do not mix 100 g values with serving values silently.
- Barcode scanning is a fast path for packaged food; manual search remains available.

6.5 Seed catalogue - Indian foods to include from day one

Vegetable dishes: aloo matar, bhindi sabzi, aloo gobi, matar paneer, palak paneer, kadai paneer, baingan bharta, mix veg, lauki sabzi, tinda sabzi, gobi matar, jeera aloo, cabbage sabzi, beans sabzi, mushroom masala

Dal / legumes: dal tadka, dal fry, moong dal, masoor dal, toor/arhar dal, chana dal, rajma, chole, kala chana, lobia, sambar, rasam, sprouts chaat

Breads / grains: roti, chapati, phulka, naan, paratha, aloo paratha, paneer paratha, poori, bhatura, steamed rice, jeera rice, brown rice, pulao, veg biryani, khichdi, daliya, oats

Breakfast / snacks: poha, upma, idli, dosa, uttapam, besan chilla, moong chilla, dhokla, khandvi, vada, pav bhaji, bhel, sev puri, samosa, kachori, makhana, roasted chana

Protein foods: paneer, tofu, soy chunks, curd/dahi, Greek yogurt, milk, eggs, egg bhurji, chicken breast, chicken curry, tandoori chicken, fish curry, grilled fish, mutton curry

Beverages / sweets: chai with milk, coffee with milk, lassi, buttermilk/chaas, coconut water, jalebi, gulab jamun, rasgulla, kheer, halwa, laddoo

Global staples: bread, pasta, noodles, tortilla, quinoa, couscous, oats, cereal, peanut butter, whey protein, yogurt, cheese, avocado, hummus, falafel, pizza, burger, sushi, tacos, burrito, grilled chicken, salmon, tuna

DATASET STRATEGY

Do not manually hard-code nutrient values for thousands of dishes. Normalize reliable source datasets, then compute composite dishes from versioned recipes and yield factors. Store source + confidence on every nutrient record.

6.6 Meal logging UX

Meal	Required behaviors
Breakfast / Lunch / Snacks / Dinner	Search foods, add quantity/serving, copy yesterday, save meal, create recipe, add notes.
Quick add	Calories/macros only for cases where food is not found.
Templates	"My usual breakfast", "Office lunch", "Post-workout shake".
Meal target view	Remaining calories/macros or simplified plate-based goals depending on user preference.
Correction	Edit serving, source or recipe; historical logs retain the nutrient version used at time of logging.

07 Training & Progression Engine

Personalized programming plus detailed session logging

7.1 Plan-generation inputs

- Athlete mode and goal weights.
- Experience/system level and training age.
- Days/week, preferred training days, max session duration.
- Equipment inventory and location.
- Exercise preferences/avoidances and pain/injury restrictions supplied by user.
- Recent sets/reps/load, RIR/RPE, completion rate, PR history and volume by muscle/movement.
- Sleep/recovery/soreness signals and bodyweight trend.
- CV proportions as a secondary aesthetic signal only.

7.2 Programming modes

Mode	Core plan logic	Examples of tracked progression
Bodybuilding	Muscle-group volume/frequency, hypertrophy rep ranges, weak-point emphasis	Load/reps, volume, RIR, set quality, muscle frequency.
Calisthenics	Skill trees + strength prerequisites + bodyweight progression	Reps, hold time, assistance band, leverage variation, weighted reps.
Athletic	Strength + power + speed + conditioning blocks	Load, velocity if available, sprint time, jump distance/height, intervals.
Aesthetic	Hypertrophy with visual-balance priorities and adherence	Body measurements, proportion trend, key lifts, volume distribution.
Hybrid	Weighted blend with interference and fatigue constraints	Mode-specific metrics plus weekly stress budget.

7.3 Session screen requirement - Day 22 example

When the athlete opens Day 22, the session page must load the planned exercises and the previous comparable performance. Each exercise presents warm-up guidance (optional), planned work sets, rest timer, technique notes and a compact logging grid.

Exercise	Set	Target	Previous	User entry	Extra fields
Bench Press	1	6-8 reps @ planned load	70 kg × 8	72.5 kg × 7	RIR 2, rest 2:30
Bench Press	2	6-8 reps	70 kg × 7	72.5 kg × 7	RIR 1
Incline DB Press	1	8-12 reps	24 kg × 11	26 kg × 9	RIR 2
Lateral Raise	1	12-20 reps	8 kg × 17	8 kg × 19	RIR 2

7.4 Exercise logging fields

- Strength/hypertrophy: load, reps, sets, RIR or RPE, rest, tempo optional.
- Bodyweight/calisthenics: reps, hold seconds, assistance, added weight, progression variation.
- Cardio: duration, distance, pace, incline/resistance, heart-rate optional.
- Sprints/athletic: distance, time, number of efforts, rest, jump/throw metric where applicable.
- All exercises: pain/discomfort toggle, notes, video attachment optional, perceived technique quality optional.

7.5 Progression rules

Rule	Example
Double progression	When all working sets reach top of rep range at target RIR, increase load next exposure.
Volume progression	Add a set only when performance is stable and recovery/adherence support it.
Regression	If repeated target misses + poor readiness occur, reduce load/sets or substitute movement.
Deload trigger	Sustained performance decline, high fatigue markers, low readiness and accumulated block duration.
Exercise substitution	Map by movement pattern, equipment, muscle target and difficulty; retain comparable history where possible.
Calisthenics skill unlock	Progress after prerequisite reps/holds and technique confidence thresholds.

7.6 Progress metrics

- Estimated 1RM trend for applicable lifts.
- Weekly hard sets / volume load by muscle and movement.
- Rep PRs, load PRs, volume PRs, skill milestones and conditioning PRs.
- Exercise consistency and missed-set reasons.
- Training adherence %, session duration and average RIR/RPE.
- Plateau detection: no meaningful improvement across a configurable exposure window.

08 Habit Tracker

Spreadsheet-like consistency tracker

8.1 UX requirement

The habit tracker should feel like a clean spreadsheet/heatmap. Rows are habits; columns are calendar days. The user can click/tap a cell to mark complete. On mobile, freeze the habit name column and horizontally scroll the dates. Month view is primary; week view is available on Today.

Feature	Behavior
Habit types	Boolean tick, count target, duration target, quantity target.
Examples	10k steps, mobility 10 min, creatine taken, no sugary drink, meditation, sleep before 11:30, 3L water.
Schedule	Daily, selected weekdays, X times/week, custom date range.
Streaks	Current, longest, monthly completion %, grace logic for non-scheduled days.
Bulk action	Copy last month, pause, archive, reorder.
Notes	Optional note on a cell/day without cluttering the matrix.

Feature	Behavior
Color intensity	Completion heatmap based on percent of target; always accessible with text/icon state.

8.2 Habit-to-outcome analytics

Once enough data exists, the dashboard may show non-causal correlations such as “Weeks with ≥ 5 sleep-target completions also had higher workout completion.” Phrase these as associations, not proof that one behavior caused another.

09 Reports & Analytics Dashboard

Weekly, monthly and block-level progress views

9.1 Dashboard layers

Layer	Questions answered
Today	What should I do next?
This week	Am I on plan? How much training/nutrition/habits are complete?
Monthly comparison	What changed vs last month?
Training block	Did performance improve across this program block?
Physique / measurements	How are weight, measurements and visual ratios trending?
Recovery	Are sleep/soreness/readiness patterns associated with training quality?

9.2 Weekly report

- Workout completion: planned vs completed sessions and sets.
- Performance: top lift/skill improvements, PRs, volume trend.
- Nutrition: days logged, average calories/macros, protein adherence if enabled, meal consistency.
- Hydration and habits: completion rates and streak changes.
- Body trend: average weekly weight (not single-day noise), circumference changes when available.
- Recovery: average sleep, soreness/readiness and any repeated pain flags.
- Next-week recommendation with a short reason: maintain, progress, reduce, substitute, or request more data.

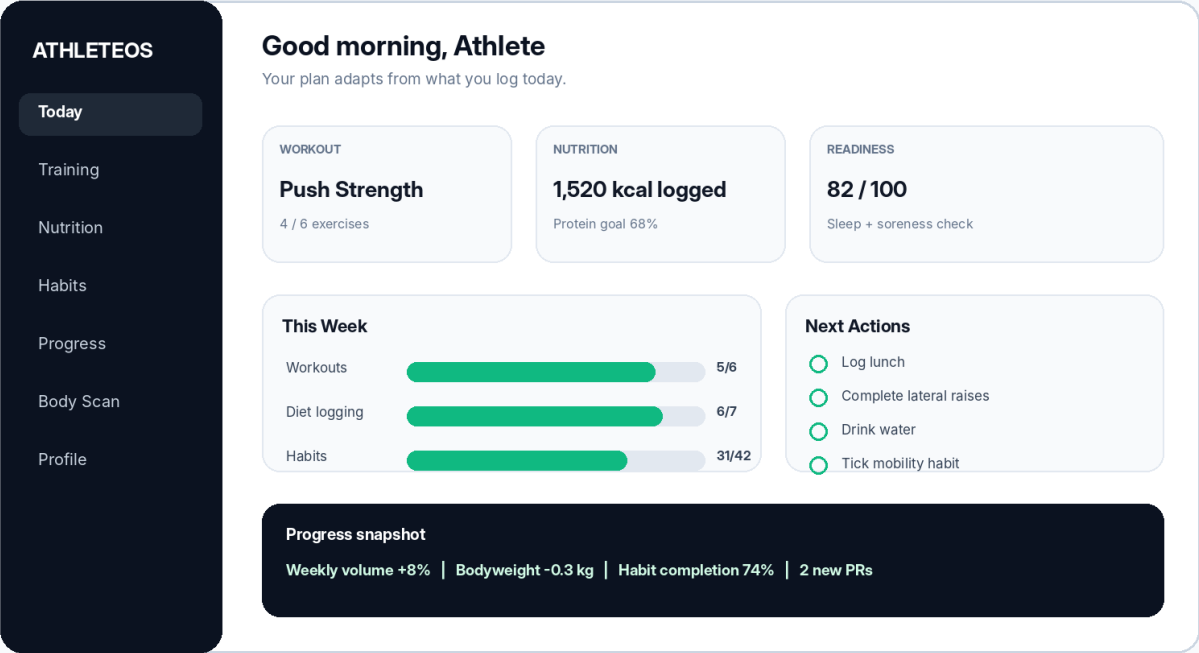
9.3 Monthly comparison

Metric group	Example comparison
Training	Workouts 18→20; volume +6%; bench estimated 1RM +3.1%
Nutrition	Logging adherence 71%→86%; average protein adherence +9 pp
Body	Weekly average weight 74.8→75.4 kg; waist -0.5 cm
CV progress	Visual shoulder/hip ratio +1.8% at high capture-confidence
Habits	Completion 63%→79%; longest mobility streak 12 days
Recovery	Average sleep 6h42→7h05; high-soreness days 7→4

9.4 Insight priority

Rank insights by actionability and confidence. Example order: pain/safety flag → adherence problem → consistent performance regression → recovery issue → plateau → successful progression → aesthetic/visual trend. Avoid flooding the user with more than 3-5 primary insights per report.

Dashboard UX Direction - Desktop / PWA



10 UI/UX Product Direction

A premium, training-first interface rather than a generic admin dashboard

10.1 Recommended design language

Area	Direction
Visual style	Performance-tech: deep charcoal/navy surfaces, clean off-white content areas, energetic green/lime accent used sparingly.
Typography	Inter/Lato/SF-style sans serif; large numeric metrics; compact labels.
Navigation	Mobile bottom bar: Today, Train, Food, Habits, Progress. Desktop sidebar adds Body Scan and Profile.
Cards	Low-radius or medium-radius cards, strong hierarchy, limited shadows, clear completion states.
Charts	One insight per chart; consistent units; week/month toggle; compare against prior period.
Motion	Fast micro-interactions: set saved, PR celebration, streak tick; avoid distracting page-load animations during workouts.
Accessibility	Light/dark mode, WCAG-conscious contrast, icon + text states, large tap targets, reduced-motion option.

10.2 Key screens

- Welcome / register / sign-in
- Athlete mode selection
- Goal & priority weighting
- Metrics onboarding
- Equipment inventory
- Diet preference + allergies
- Body scan capture & results
- Plan preview
- Today dashboard

- Workout session logger
- Exercise detail/history
- Food search + meal composer
- Recipe builder
- Habit spreadsheet
- Weekly report
- Monthly comparison
- Progress photos / body metrics
- Settings / privacy / export
- Admin console

10.3 Workout UX details

- Keep the current exercise, current set and previous performance above the fold.
- Large one-handed +/- load and reps controls; numeric keypad on mobile.
- Rest timer begins automatically after save but can be disabled.
- Swipe or tap to move to next set; never require navigating back to a separate history screen.
- PR badge appears after save but does not block logging.
- Offline-safe session logging with sync when connection returns is a high-value P1 requirement.

10.4 Food UX details

- Search bar opens directly to recents/favorites, not a blank state.
- Indian household measures displayed alongside grams; users can customize a personal katori/cup size later.
- Quick copy: yesterday's breakfast, last Monday's lunch, saved meal template.
- Clearly mark recipe-derived and user-entered nutrition estimates.
- Barcode scanner can be launched from the same search surface.

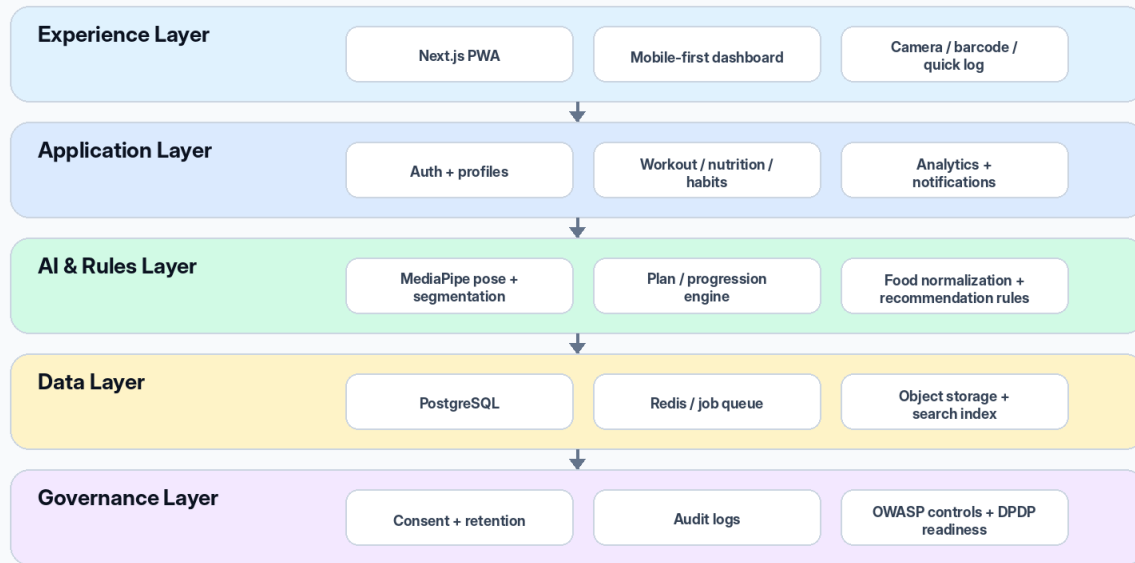
11 Data Model & System Architecture

Recommended technical shape for a scalable first version

11.1 Recommended stack

Layer	Recommendation	Why
Frontend	Next.js + TypeScript PWA	Strong responsive app model, server rendering where useful, installable mobile experience.
UI	Tailwind CSS + accessible component primitives	Rapid consistent design system and responsive states.
Backend	FastAPI (Python)	Natural fit for ML/CV services, async APIs and data processing.
Primary database	PostgreSQL	Relational integrity for users, plans, logs, metrics, foods and analytics.
Cache/jobs	Redis + worker queue	Plan generation, report calculations, notifications, data sync jobs.
Object storage	S3-compatible encrypted storage	Optional progress photos/videos; signed URLs and lifecycle rules.
Food search	PostgreSQL trigram initially; Typesense/Meilisearch at scale	Fast typo-tolerant aliases and ranking.
CV execution	MediaPipe in browser where possible; server fallback	Privacy, latency and lower image-storage requirement.
Analytics	Event pipeline + warehouse later	Product analytics separated from athlete operational database.

Recommended Product Architecture



11.2 Core entities

Entity	Purpose
users	identity, locale, auth status
athlete_profiles	goals, mode weights, level, equipment, preferences
body_metrics	timestamped weight/circumferences/body-derived features
body_scans	consent, capture metadata, derived landmarks/ratios, optional media reference
training_plans	versioned generated program + rationale
workout_days	planned day/session metadata
exercises	movement taxonomy, muscles, equipment, progression family
workout_sets	planned vs performed load/reps/RIR/etc.
foods	normalized food entities + source/confidence
food_aliases	search synonyms and transliterations
recipes	ingredient quantities, yield, nutrition calculation version
meal_logs	meal → food/recipe serving entries
habits	definition/schedule
habit_logs	per-day completion/value
readiness_logs	sleep, soreness, energy, resting HR optional
insights	generated insight, confidence, rationale, action
notifications	scheduled/push/email state
consents	purpose, version, timestamp, withdrawal

11.3 API groups

- /auth, /profile, /metrics, /body-scan, /plans, /workouts, /exercise-library, /foods, /recipes, /meals, /habits, /readiness, /reports, /insights, /notifications, /admin.
- Every write API should be idempotent where practical for mobile/offline retry behavior.
- Version generated plans and reports so historical results remain reproducible after algorithms change.

12 Security, Privacy & Safety

Especially important because progress photos and body metrics are personal data

12.1 Privacy requirements

- Separate consent for body-photo processing/storage from basic account consent.
- Offer “process and discard image” mode where only landmarks/derived metrics are retained.
- Provide raw-photo retention controls and automatic deletion policy.
- Allow export and deletion of athlete data and media.
- Do not use user progress photos to train future models without an explicit, separate opt-in.
- Minimize demographic inputs; only collect fields required for a chosen personalization feature.
- Maintain consent version, purpose, timestamp and withdrawal audit trail.
- Design for India DPDP Act/Rules readiness and obtain legal review before production launch. [5]

12.2 Application security

Use OWASP ASVS as the security verification baseline for authentication, authorization, API design, input handling and secure file upload controls. [6]

Area	Requirement
Authentication	Strong password hashing, session rotation, rate limiting, optional MFA/passkeys, suspicious-login controls.
Authorization	Object-level checks on every athlete resource; no predictable media access.
Uploads	Validate content type by file signature, size limits, image re-encoding, malware scanning where applicable, random object names.
Storage	Encryption in transit and at rest; signed short-lived media URLs.
API	Schema validation, rate limits, CSRF strategy where relevant, CORS allowlist, abuse detection.
Secrets	Never expose API keys in client bundles; rotate keys; environment/secret manager.
Logging	No body images or full sensitive payloads in logs; audit access to sensitive media.
Backups	Encrypted backups, tested restore, deletion propagation policy.

12.3 Fitness safety guardrails

- Pain/injury flags should stop automatic progression and suggest professional evaluation for concerning or persistent issues rather than diagnosing.
- Nutrition and hydration outputs must be framed as estimates/targets, not medical prescriptions.
- Users with pregnancy, eating disorder history, significant medical conditions or medications should be routed to professional guidance instead of aggressive automated goals.
- Advanced intensity techniques (forced reps, extreme deficits, high-risk skill progressions) should not be auto-prescribed to beginners.
- Show exercise demos and key safety cues; allow user to replace any uncomfortable exercise.

13 Non-Functional Requirements

Quality expectations that make the product usable in the gym

Category	Target / requirement
Performance	Today dashboard interactive quickly on normal 4G; set-save response perceived <300 ms via optimistic UI.
Reliability	Workout logs never lost; local draft/offline queue for active session.
Availability	Target 99.9% for production core APIs after MVP stabilization.
Scalability	Stateless app services; background reports; cache food search and read-heavy catalog data.
Mobile UX	Primary workflows usable at 360 px width; tap targets suitable for sweaty/one-handed gym use.
Accessibility	Keyboard navigation on desktop, screen-reader labels, contrast-safe states, reduced motion.
Localization	Metric/imperial; architecture ready for English + Hindi/transliteration and additional languages.
Observability	Structured logs, traces, error monitoring, job monitoring, model/version metadata.

Category	Target / requirement
Data quality	Every nutrition value and CV metric carries source/version/confidence metadata.
Explainability	Plan changes and insights include a short human-readable rationale.

14 Admin & Operations

Tools required to keep data and recommendations trustworthy

14.1 Admin console modules

- User/support lookup with strict role-based access and audited actions.
- Exercise library CMS: muscles, equipment, instructions, media, substitutions, progression families.
- Food catalog tools: merge duplicates, aliases, recipe verification, source version, nutrition confidence.
- Program template editor for each athlete mode/level.
- Model configuration: rule thresholds, model version, feature flags, rollback.
- Content management: onboarding tips, exercise safety copy, notifications, report explanations.
- Abuse/security dashboard: failed logins, suspicious uploads, rate-limit events.
- Data import jobs for IFCT/FDC/OFF normalization with staging + validation before publish.

14.2 Model and rule governance

Artifact	Versioning requirement
CV model / landmarks	model name, asset version, threshold config
Derived ratios	formula version + capture protocol version
Training planner	planner version + program template version + rationale snapshot
Nutrition catalog	source dataset version + normalization revision
Recipe calculation	ingredient versions + yield factor + recipe version
Reports	aggregation version so historical reports are reproducible

15 Product Metrics & Analytics

How to know whether AthleteOS is actually helping

15.1 North-star metric

WEEKLY COMPLETED PLANNED ACTIONS

Percentage of planned training, nutrition-logging and habit actions completed by active athletes, weighted toward training execution. This measures whether the product turns plans into behavior.

15.2 Supporting KPIs

Area	KPIs
Activation	Onboarding completion, first plan generated, first workout completed, first meal logged.
Retention	D1/D7/D30 retained athletes; weekly active loggers; 4-week plan continuation.
Training	Session completion, set logging completion, PR rate, skipped-exercise rate, plan override rate.
Nutrition	Days logged/week, food search success, no-result rate, recipe creation, repeat/favorite usage.
Habits	Habit completion %, streak continuation, habit creation vs abandonment.
CV	Scan completion, recapture rate, quality-score distribution, raw-photo storage opt-in.
Insights	Insight open rate, action taken, "helpful" feedback, recommendation override.

Area	KPIs
Quality	Crash/error rate, lost-log incidents, support tickets, nutrition correction rate.

16 Release Plan & Priorities

Build the behavior loop first; add advanced AI after sufficient data exists

Phase	Scope	Exit criteria
Phase 0 - Prototype	Clickable UI, onboarding, workout logger, habit grid, sample dashboard	10-20 target users can complete core flows without guidance.
Phase 1 - MVP	Auth, profiles, 4 athlete modes, rule-based plan generation, exercise library, set logging, streaks, habit grid, food search + recipes, weekly/monthly dashboard	Stable daily use; no lost logs; food search works well for common Indian foods.
Phase 2 - Smart Personalization	Readiness, adaptive progression, plateau detection, improved reports, MediaPipe body scan, barcode packaged foods, PWA offline workout	Recommendations explain changes and pass coach review benchmarks.
Phase 3 - Scale	Wearables, coach mode, advanced body-scan comparison, notifications, subscriptions, data warehouse, A/B testing	Retention and unit economics support growth.
Phase 4 - Advanced AI	Exercise form analysis for selected movements, multimodal food/photo assistance, personalized coach chat, learned recommendation ranking	Only after validated datasets, safety evaluation and monitoring are in place.

16.1 Suggested MVP priority

- P0: auth/onboarding, athlete profile, plan templates, workout set logger, exercise history, streaks, habit matrix, meal logging, Indian food aliases, weekly/monthly dashboard, data export/delete.
- P1: body scan, barcode, readiness, adaptive progression, offline PWA, notifications, richer reports.
- P2: wearables, coach portal, social/community, advanced CV form coaching, AI chat, subscriptions.

17 Acceptance Criteria

Definition of “done” for the first production-ready version

Capability	Acceptance criterion
Onboarding	A new athlete can register, enter baseline metrics, choose a mode/goal/level/equipment and receive a valid first-week plan.
Workout logging	An athlete can open any plan day (e.g., Day 22), see planned and previous performance, log every set, leave and resume without losing data.
Progression	The system can propose next-load/rep/set changes using deterministic rules and display why.
Food search	Queries for common Indian foods and variants such as “aloo matar”, “bhindi”, “rajma chawal”, “poha” return useful results or recipe options.
Meal history	Daily breakfast/lunch/snacks/dinner totals can be edited retrospectively and reports update correctly.
Habit grid	User can create recurring habits, tick a month matrix and see streak/completion statistics.
Reports	Week and month comparisons show training adherence, progress metrics, nutrition/habit adherence and body trend when data exists.
Body scan	Consent is explicit; low-quality images are rejected; derived metrics include confidence; user can choose not to store raw images.
Privacy	User can download data, delete account/media and revoke photo consent.
Security	Authorization tests prevent cross-user access; upload and

Capability	Acceptance criterion
	authentication controls meet the defined security checklist.
Responsiveness	Core logging flows function on modern desktop and mobile browsers.
Explainability	Any automatic plan change has a plain-language rationale and can be overridden by the user.

18 Risks & Mitigations

The highest-risk assumptions in this product

Risk	Impact	Mitigation
Photo-based body analysis overpromises precision	Trust/safety damage	Limit to pose/proportions/progress; confidence labels; no clinical claims.
Indian dish nutrition varies by recipe/oil	Incorrect calorie estimates	Recipe variants, serving calibration, source/confidence, user recipe customization.
Too much onboarding	Drop-off before value	Progressive onboarding; require only minimum fields; scan and circumference optional.
Workout plan feels generic	Low retention	Use schedule/equipment/history and adaptation; explain decisions; allow exercise swaps.
Manual logging fatigue	Abandonment	Recents, copy, favorites, templates, barcode, one-tap set reuse.
AI changes plan too frequently	Confusion / loss of trust	Minimum evidence windows, bounded changes, user lock and "why changed".
Sensitive media breach	Severe privacy harm	On-device processing, opt-in storage, signed URLs, deletion, least privilege and audit logs.
Nutrition/fitness recommendations unsafe for special populations	Health risk	Screening questions, conservative boundaries, professional-referral messaging.

19 Future Expansion

Features to add only after the core loop is strong

Future module	Description
AI exercise form coach	Camera-based rep counting and technique cues for a limited validated exercise set.
Wearables	Apple Health / Health Connect / Garmin/Fitbit where APIs permit; steps, sleep, HR, workouts.
Coach portal	Coach can assign plans, review sessions, comment and monitor athlete adherence with consent.
Gym mode	QR/NFC equipment identification, plate calculator, rest timer on lock screen, live gym session UI.
Food photo assist	Image-assisted candidate foods + portion prompt; user confirms before logging.
Community	Private challenges, friend streaks, leaderboards with opt-in and anti-comparison safeguards.
Periodization studio	Advanced block planning, meet/competition prep, peaking and deload controls.
Recovery intelligence	Personal baseline models using sleep, soreness, HR/HRV when reliable wearable data exists.
Commerce	Optional coach subscriptions, premium analytics, program marketplace; keep recommendation integrity separate from sponsored content.

20 Reference Sources

Primary/official sources used for technical and governance decisions

Source	URL	Used for
[1] Google AI Edge - MediaPipe Pose Landmarker for Web	https://developers.google.com/edge/mediapipe/solutions/vision/pose_landmarker/web_js	Pose landmarks, world coordinates, optional segmentation mask and web implementation

Source	URL	Used for
[2] ICMR-National Institute of Nutrition - Indian Food Composition Tables 2017	https://www.nin.res.in/ebooks/IFCT2017.pdf	Indian foundational food composition source; NIN describes the 2017 tables as covering 528 key foods and extensive components.
[3] USDA FoodData Central API Guide	https://fdc.nal.usda.gov/api-guide	REST food search/details API and public-domain/CC0 licensing information.
[4] Open Food Facts API Documentation	https://openfoodfacts.github.io/openfoodfacts-server/api/tutorial-off-api/	Packaged-food product/barcode API and nutrition data access.
[5] Ministry of Electronics and Information Technology - Digital Personal Data Protection Rules 2025	https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWVa	India privacy-regulatory reference for production legal review.
[6] OWASP Application Security Verification Standard (ASVS)	https://owasp.org/www-project-application-security-verification-standard/	Security verification baseline for web application controls.

FINAL PRODUCT RECOMMENDATION

Build Phase 1 as a polished training + food + habit execution system. Ship CV as an optional assistive feature in Phase 2, not as the product's main promise. The strongest moat will come from high-quality longitudinal athlete data and adaptive recommendations that users can understand and trust.

A Appendix - Key Calculation & Recommendation Notes

Implementation guidance for later TRD/model specification

A.1 Energy / nutrition targets

The product may estimate maintenance energy using a validated equation and activity inputs, but all results should be labeled estimates and adjusted from multi-week observed weight trend and user goal. Avoid overreacting to single-day weight changes. Nutrition targets should support configurable coach/dietitian overrides and special-population guardrails.

A.2 Readiness score

Use a transparent weighted score from sleep vs personal baseline, soreness, energy/motivation, prior-session RPE, optional resting heart rate and recent performance. Do not pretend the score is medical. The system should show the top 1-2 factors affecting the score and cap how much it can change a workout.

A.3 Plan adaptation precedence

- 1) Safety/pain constraints
- 2) Equipment/schedule constraints
- 3) Workout adherence and actual performance
- 4) Recovery/readiness
- 5) Goal priorities and muscle/skill emphasis
- 6) Bodyweight/nutrition trend
- 7) CV visual-proportion signal

A.4 Recommended next artifacts

- Technical Requirements Document (TRD) with services, APIs, database schema and deployment plan.
- Normalized PostgreSQL schema + ERD.
- Exercise library dataset and progression-tree specification.
- Food catalog ingestion pipeline + alias dataset and recipe normalization rules.
- UI/UX wireframes and design system.
- AI/CV model evaluation plan and body-scan capture prototype.
- MVP sprint plan with epics, user stories and acceptance tests.